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Permutations of $[2n]$ with an increasing subsequence of length n: a proof of the Kauers–Koutschan recurrence

Conjecture 17 of Kauers–Koutschan (OEIS `A269021`) proved, via the discrete Bessel kernel and a Wilf–Zeilberger certificate

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Abstract

Let $a(n)$ be the number of permutations of $\{1,\ldots,2n\}$ that contain an increasing subsequence of length n (OEIS `A269021`). From 42 known terms, Kauers and Koutschan guessed a linear recurrence for $a(n)$ of order 4 with polynomial coefficients of degree 21, and listed it as an open conjecture (their Conjecture 17): no proof was available because no finite, a priori representation of $a(n)$ was known — the classical D-finiteness results for pattern containment fix the pattern length, while here the pattern length n is coupled to the permutation length $2n$. We prove the conjectured recurrence. The key is the exact representation

$$a(n) \;=\; (2n)!^2 \sum_{\rho=n}^{2n} \frac{(-1)^{\rho-n}\binom{2\rho-2}{\rho-n}}{(\rho!)^2\,(2n-\rho)!}\,,$$

which follows from the Robinson–Schensted correspondence, the Borodin–Okounkov–Olshanski formula for the one-point correlation function of the poissonized Plancherel measure (the discrete Bessel kernel), and the elementary observation that a partition of $2n$ has at most one row of length at least n, so the relevant inclusion–exclusion truncates after one term. The summand is a proper hypergeometric term vanishing outside $n \le \rho \le 2n$, and Zeilberger's algorithm returns *exactly* the Kauers–Koutschan operator as a telescoper for this sum; we exhibit the rational certificate and prove the two resulting rational-function identities deterministically (exact evaluation on integer grids exceeding explicit degree bounds). As corollaries, $a(n)/(2n)!^2$ is P-recursive — to our knowledge the first D-finiteness proof for this coupled diagonal — and the same method gives a single-sum formula for the number of permutations of $[2n]$ whose longest increasing subsequence has any prescribed length $\ge n$. All computations are reproducible from one self-contained script.

MSC 2020: 05A15, 05A05 (primary); 05E10, 33F10, 68W30 (secondary). **Keywords:** longest increasing subsequence; Robinson–Schensted; Plancherel measure; discrete Bessel kernel; creative telescoping; Wilf–Zeilberger certificate; D-finite; OEIS A269021.

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Status & provenance. This is an unrefereed, computer-assisted preprint. The certificate was found by the `zeilberger` package of Maxima and every step of the verification is mechanical (Section 4); the two identities it rests on are proved by exact evaluation beyond explicit degree bounds, which a reader can re-run in ∼10 seconds from the shipped script, whose data are part of the proof object (Section 4). Two ingredients are *cited, not re-derived*: the Robinson–Schensted correspondence (Schensted [3]), and the Borodin–Okounkov–Olshanski one-point formula with its kernel series ([2], Theorem 2 and Proposition 2.6, eq. (2.14), arXiv v2 numbering; the statements used are quoted in Section 2, and were cross-checked numerically in exact arithmetic at 34 lattice points during the research phase, with the shipped script re-verifying all downstream consequences). **Novelty:** the ingredients are classical and the representation (2) is a natural consequence of [2] available since 2000; we found no prior statement of it or of the resulting proof, but make **no priority claim** and would welcome pointers to prior appearances. AI-assisted by Claude under the Periapsis Architecture (acknowledgements).

1 Introduction

In their automated study of possibly-D-finite sequences in the OEIS, Kauers and Koutschan [1] examined the sequence `A269021`,

$$a(n) \;=\; \#\{\sigma \in S_{2n} \;:\; \sigma \text{ contains the pattern } 12\cdots n\} \;=\; \#\{\sigma \in S_{2n} : \mathrm{LIS}(\sigma) \geq n\},$$

where LIS denotes the length of the longest increasing subsequence:

$$1,\ 2,\ 23,\ 588,\ 24553,\ 1438112,\ 108469917,\ \ldots \qquad (n \geq 0).$$

From the 42 terms then in the OEIS they detected a linear recurrence of order 4 and degree 21,

$$c_0(n)\,a(n) + c_1(n)\,a(n{+}1) + c_2(n)\,a(n{+}2) + c_3(n)\,a(n{+}3) + c_4(n)\,a(n{+}4) \;=\; 0, \tag{1}$$

with explicit $c_i \in \mathbb{Z}[n]$, $\deg c_i \leq 21$ (the coefficients are reproduced in Appendix A); equivalently, in the normalization $\tilde{a}_n = a(n)/(2n)!^2$ used in their paper, this is their Conjecture 17. They marked the case **O** — open: neither the recurrence nor even the D-finiteness of the sequence was proved. They are explicit about the obstruction [1, §6]:

> *"It is known [Gessel; Bostan et al.] that for every fixed k, the number of permutations of length n avoiding the pattern $123\cdots k$ is D-finite as a sequence in n. However, this result has no immediate implications on sequences we obtain when n and k are coupled."*

Indeed, for *fixed* k the counting sequence is governed by Gessel's Bessel–Toeplitz determinant [4], but the determinant's size grows with k; on the coupled diagonal $k = n$, $m = 2n$ no finite a priori representation — a finite sum or integral of hypergeometric type to which creative telescoping could be applied — was on record. That missing representation is the entire content of the present note; once it is found, the rest is the standard Wilf–Zeilberger method, and the certificate turns out to land exactly on the guessed operator.

Theorem 1. *The recurrence* (1) *holds for all integers $n \geq 0$ with the exact coefficients $c_0, \ldots, c_4$ recorded by Kauers and Koutschan. Equivalently, Conjecture 17 of [1] holds. In particular $\tilde{a}_n = a(n)/(2n)!^2$ is P-recursive.*

The proof has two independent halves:

1. **A proven representation (Section 2).** We show

$$a(n) \;=\; \sum_{\rho\in\mathbb{Z}} t(n,\rho), \qquad t(n,\rho) := \frac{(-1)^{\rho-n}\binom{2\rho-2}{\rho-n}\,(2n)!^2}{(\rho!)^2\,(2n-\rho)!}\,, \tag{2}$$

   the summand vanishing outside $n \le \rho \le 2n$. This rests on the Robinson–Schensted correspondence, on the Borodin–Okounkov–Olshanski (BOO) formula for the one-point correlation function of the poissonized Plancherel measure, and on the *one-row truncation*, a one-line combinatorial observation: a partition of $2n$ has at most one row of length $\ge n$.

2. **A Wilf–Zeilberger certificate (Section 3).** The term $t(n,\rho)$ is proper hypergeometric, and the Kauers–Koutschan operator $L = \sum_{i=0}^{4} c_i(n)N^i$ (where N is the shift $n \mapsto n+1$) telescopes it: we exhibit an explicit rational function $\widetilde{R}(n,\rho)$ with $L\,t = G(n,\rho{+}1) - G(n,\rho)$ for a closed-form G built from $\widetilde{R}$, and prove the two rational-function identities this requires by exact evaluation on integer grids exceeding explicit degree bounds — a deterministic, finite proof. Summing over ρ then gives $L\,a = 0$ for all $n \ge 5$; the finitely many initial cases are checked directly from (2).

1.1 Why a term-match is not a proof, and what is genuinely needed

The 42-term b-file cannot prove (1): the recurrence has $5\cdot 22 = 110$ unknown coefficients (109 degrees of freedom after overall scaling), and any candidate operator fitted to data — or any second operator obtained from data by closure properties — begs the question, since exhibiting *any* bounded-order annihilator of $a(n)$ is essentially equivalent to the D-finiteness assertion being proved. (Kauers and Koutschan of course knew this; it is why they published the recurrence as a conjecture.) What breaks the circle is a representation of $a(n)$ that is proved *before* any recurrence enters: (2) is derived from published theorems with no reference to (1), and the conjectured operator then enters only as the *candidate telescoper* whose certificate is verified symbolically. This is the standard architecture of a Wilf–Zeilberger proof [5, 7], and it is the exact route Kauers–Koutschan use for the conjectures in their paper that they *could* prove — what was missing for `A269021` was only the representation.

2 The representation

Throughout, $\lambda = (\lambda_1 \ge \lambda_2 \ge \cdots)$ is a partition, $|\lambda|$ its size, f^λ the number of standard Young tableaux of shape λ.

2.1 Ingredients

(i) Robinson–Schensted. $\mathrm{LIS}(\sigma) = \lambda_1(\mathrm{RS}(\sigma))$ and the correspondence is a bijection $S_m \to \bigcup_{\lambda\vdash m}\mathrm{SYT}(\lambda)^2$ [3], so

$$a(n) \;=\; \sum_{\substack{\lambda\vdash 2n\\ \lambda_1\ge n}} (f^\lambda)^2. \tag{3}$$

(ii) The one-row truncation. For a partition λ define its *descent set*

$$\mathcal{D}(\lambda) \;=\; \{\lambda_i - i \;:\; i\ge 1\} \;\subset\; \mathbb{Z}$$

(the convention of [2, §1.2]; the map $i \mapsto \lambda_i - i$ is strictly decreasing, so each value is attained at most once).

Lemma 2 (one-row truncation)**.** *Let $\lambda \vdash 2n$ and $x \geq n-1$ an integer. If $x \in \mathcal{D}(\lambda)$, i.e. $x = \lambda_i - i$ for some i, then $i = 1$. Consequently*

$$\#(\mathcal{D}(\lambda) \cap \mathbb{Z}_{\geq n-1}) \;=\; \begin{cases} 1, & \lambda_1 \geq n \quad (\textit{the element } \lambda_1 - 1), \\ 0, & \lambda_1 \leq n-1. \end{cases}$$

Proof. If $\lambda_i - i = x \geq n-1$ with $i \geq 2$ then $\lambda_2 \geq \lambda_i = x + i \geq n+1$, so $|\lambda| \geq \lambda_1 + \lambda_2 \geq 2\lambda_2 \geq 2n+2 > 2n$, a contradiction. For $i = 1$, $\lambda_1 - 1 \geq n-1 \iff \lambda_1 \geq n$. $\square$

Combining with (3),

$$a(n) \;=\; \sum_{\lambda \vdash 2n} (f^\lambda)^2 \cdot \#(\mathcal{D}(\lambda) \cap \mathbb{Z}_{\geq n-1}). \tag{4}$$

(iii) The BOO one-point function. For $\theta > 0$ the *poissonized Plancherel measure* on the set of all partitions is

$$M^\theta(\lambda) \;=\; e^{-\theta}\, \theta^{|\lambda|} \Big(\frac{f^\lambda}{|\lambda|!}\Big)^2.$$

Borodin, Okounkov and Olshanski prove ([2], the second theorem of their introduction — Theorem 2 in the arXiv numbering) that the correlation functions of $\mathcal{D}(\lambda)$ under M^θ are determinantal with the *discrete Bessel kernel*; we need only the one-point case: for every $x \in \mathbb{Z}$,

$$M^\theta(\{\lambda : x \in \mathcal{D}(\lambda)\}) \;=\; \mathcal{J}(x,x;\theta), \qquad \mathcal{J}(x,y;\theta) = \sqrt{\theta}\, \frac{J_x J_{y+1} - J_{x+1} J_y}{x-y}, \tag{5}$$

where $J_x = J_x(2\sqrt{\theta})$ is the Bessel function (the diagonal is the usual limit). We use the denominator-free series form ([2], Proposition 2.6, eq. (2.14) in the arXiv v2 numbering):

$$\mathcal{J}(x,y;\theta) \;=\; \sum_{m \geq 0} (-1)^m \frac{(x+y+m+2)_m}{\Gamma(x+m+2)\,\Gamma(y+m+2)} \cdot \frac{\theta^{\frac{x+y}{2}+m+1}}{m!}\,, \tag{6}$$

where $(z)_m = z(z+1)\cdots(z+m-1)$ denotes the rising factorial. (An equivalent expansion $\mathcal{J}(x,y) = \sum_{s \geq 1} J_{x+s} J_{y+s}$ is also derived there; we will not need it.)

2.2 The formula

Proposition 3. *For all $n \geq 1$,*

$$a(n) \;=\; (2n)!^2 \sum_{\rho=n}^{2n} \frac{(-1)^{\rho-n}\binom{2\rho-2}{\rho-n}}{(\rho!)^2\,(2n-\rho)!}\,, \tag{7}$$

and, refining by the exact value of the longest increasing subsequence, for every $j \geq 0$,

$$\#\{\sigma \in S_{2n} : \mathrm{LIS}(\sigma) = n+j\} \;=\; (2n)!^2 \sum_{\rho=n+j}^{2n} \frac{(-1)^{\rho-n-j}\binom{2\rho-1}{\rho-n-j}}{(\rho!)^2\,(2n-\rho)!}\,. \tag{8}$$

Proof. Fix an integer $x \geq n-1$. Multiplying (5) by e^θ gives

$$\sum_\lambda \theta^{|\lambda|} \Big(\frac{f^\lambda}{|\lambda|!}\Big)^2 [\,x \in \mathcal{D}(\lambda)\,] \;=\; e^\theta\, \mathcal{J}(x,x;\theta) \qquad (\theta > 0).$$

Both sides are entire functions of θ: the left side is a power series whose θ^m coefficient is at most $\sum_{\lambda\vdash m}(f^\lambda/m!)^2 = 1/m!$, and the right side is e^θ times the everywhere-convergent series (6). Agreement on $(0,\infty)$ therefore forces agreement of all power-series coefficients. Extracting $[\theta^{2n}]$ (only $|\lambda| = 2n$ contributes on the left):

$$\frac{1}{(2n)!^2}\sum_{\lambda\vdash 2n}(f^\lambda)^2\,[\,x\in\mathcal{D}(\lambda)\,] \;=\; [\theta^{2n}]\;e^\theta\,\mathcal{J}(x,x;\theta). \tag{9}$$

On the diagonal $x = y$, with $x \geq 0$ an integer, (6) reads

$$\mathcal{J}(x,x;\theta) = \sum_{m\geq 0}(-1)^m\frac{(2x+m+2)_m}{((x+m+1)!)^2\,m!}\,\theta^{x+m+1}.$$

Summing (9) over all integers $x \geq n-1$ and using (4):

$$\frac{a(n)}{(2n)!^2} \;=\; [\theta^{2n}]\;e^\theta\sum_{x\geq n-1}\mathcal{J}(x,x;\theta).$$

For this coefficient extraction the sum over x is finite: the diagonal series (6) starts at θ^{x+1}, so every $x \geq 2n$ contributes nothing to $[\theta^{2n}]$, and the interchange of the sum over x with the coefficient extraction is the interchange of a finite sum. Reindex the double series over (x,m) by $\rho = x+m+1$ and $k = x+1$ (so k runs over $n \leq k \leq \rho$ and $m = \rho - k$). Since $(2x+m+2)_m = (k+\rho)(k+\rho+1)\cdots(2\rho-1) = \frac{(2\rho-1)!}{(\rho+k-1)!}$ and $(x+m+1)! = \rho!$,

$$\sum_{x\geq n-1}\mathcal{J}(x,x;\theta) \;=\; \sum_{\rho\geq n}\frac{\theta^\rho}{(\rho!)^2}\sum_{k=n}^{\rho}(-1)^{\rho-k}\binom{2\rho-1}{\rho-k}.$$

The inner alternating partial sum collapses by the elementary identity

$$\sum_{d=0}^{D}(-1)^d\binom{M}{d} \;=\; (-1)^D\binom{M-1}{D} \qquad (M\geq 1,\;D\geq 0), \tag{10}$$

(immediate by induction on D from Pascal's rule), with $M = 2\rho-1$, $d = \rho-k$, $D = \rho-n$:

$$\sum_{k=n}^{\rho}(-1)^{\rho-k}\binom{2\rho-1}{\rho-k} \;=\; (-1)^{\rho-n}\binom{2\rho-2}{\rho-n}.$$

Finally $[\theta^{2n}]e^\theta\theta^\rho = 1/(2n-\rho)!$ (interpreted as 0 for $\rho > 2n$), giving (7). For (8), use the single point $x = n+j-1$ in (9) instead of the sum over x: by Lemma 2, $x \in \mathcal{D}(\lambda) \iff \lambda_1 = n+j$ for $\lambda \vdash 2n$, and the same reindexing (now with k fixed at $n+j$, without the collapse step) gives the stated single sum with $\binom{2\rho-1}{\rho-n-j}$. $\square$

Remark 4. Proposition 3 is exact, not asymptotic, and it is heavily over-verified in the reproducibility script: (7) agrees with a brute-force count over *all* $(2n)!$ permutations for $n \leq 3$, with the hook-length sum (3) for $n \leq 10$, and with the 16 OEIS-listed terms; during the research run it was checked against 81 independently computed terms ($n \leq 80$) and (8) against the full exact array of LIS distributions for $n \leq 26$ (494 values), with no discrepancies.

Remark 5. The right side of (8) makes sense for all $j \geq 0$ and vanishes for $j > n$, as it must. Identity (8) is a closed single-sum formula for the upper half of the distribution of LIS on S_{2n} — the regime where at most one row of the Plancherel shape can be long, so the Fredholm/inclusion–exclusion expansion of the gap probability truncates at its first term. We are not aware of it appearing in the literature, though all ingredients date to 2000 or earlier.

3 The Wilf–Zeilberger certificate

Write $t(n,\rho)$ for the summand of (2), extended to all integers ρ by its natural conventions: $\binom{2\rho-2}{\rho-n}=0$ for $\rho<n$ (negative lower index) and $1/(2n-\rho)!=0$ for $\rho>2n$. Thus $t(n,\cdot)$ is supported on $[n,2n]$ and $a(n)=\sum_{\rho\in\mathbb{Z}} t(n,\rho)$, a finite sum. Let

$$L \;=\; c_0(n)+c_1(n)N+c_2(n)N^2+c_3(n)N^3+c_4(n)N^4$$

be the Kauers–Koutschan operator (Appendix A), acting on the n variable, and set

$$(Lt)(n,\rho) \;=\; \sum_{i=0}^{4} c_i(n)\,t(n+i,\rho),$$

a combination supported on $n\le\rho\le 2n+8$.

3.1 Discovery, and what is actually proved

Running Zeilberger's algorithm (Maxima's `zeilberger` package [9]) on the hypergeometric term $t(n,\rho)$ returns a telescoper of order 4 — and its coefficients agree *exactly*, polynomial for polynomial, with the guessed $c_0,\dots,c_4$. (What we use, and prove below, is only that the operator *is* a telescoper; minimality is neither claimed nor needed.) The algorithm also returns a rational certificate. We stress the logical role of this step: it is *discovery only*. Nothing below depends on the correctness of Maxima; the certificate is re-stated in closed form and the identities it must satisfy are proved independently and deterministically.

3.2 The certificate and the two identities

All the rational functions below arise from ratios of t; the elementary factorial manipulations are stated as a lemma.

Lemma 6. *For integers ν,ρ in the indicated ranges (all denominators nonzero),*

$$\frac{t(\nu,\rho+1)}{t(\nu,\rho)}=\frac{-(2\rho)(2\rho{-}1)(2\nu{-}\rho)}{(\rho{+}1{-}\nu)(\rho{+}\nu{-}1)(\rho{+}1)^2},\qquad \frac{t(\nu{+}1,\rho)}{t(\nu,\rho)}=\frac{-(2\nu{+}1)^2(2\nu{+}2)^2(\rho{-}\nu)}{(\rho{+}\nu{-}1)(2\nu{+}2{-}\rho)(2\nu{+}1{-}\rho)}.$$

Proof. Direct computation from the definition of t; e.g. for the second, $\binom{2\rho-2}{\rho-\nu-1}/\binom{2\rho-2}{\rho-\nu}=\frac{\rho-\nu}{\rho+\nu-1}$ and $(2\nu{+}2)!^2/(2\nu)!^2=(2\nu{+}1)^2(2\nu{+}2)^2$, while $(2\nu{-}\rho)!/(2\nu{+}2{-}\rho)!=1/\big((2\nu{+}2{-}\rho)(2\nu{+}1{-}\rho)\big)$, and the sign alternates once. (The script re-checks both formulas against direct factorial ratios at every interior lattice point of several rows.) $\square$

We work in the basis of the largest shift. Define, for $0\le i\le 4$,

$$\sigma_i(n,\rho)\;:=\;\prod_{k=i}^{3}\frac{-(\rho{+}n{+}k{-}1)\,(2n{+}2k{+}2{-}\rho)\,(2n{+}2k{+}1{-}\rho)}{(2n{+}2k{+}1)^2(2n{+}2k{+}2)^2\,(\rho{-}n{-}k)}\,,\tag{11}$$

$$\tilde r(n,\rho)\;:=\;\frac{-(2\rho)(2\rho{-}1)(2n{+}8{-}\rho)}{(\rho{-}n{-}3)(\rho{+}n{+}3)(\rho{+}1)^2}\,,\tag{12}$$

so that by Lemma 6 (iterated), wherever the terms are nonzero, $\sigma_i=t(n{+}i,\rho)/t(n{+}4,\rho)$ and $\tilde r=t(n{+}4,\rho{+}1)/t(n{+}4,\rho)$ (in particular $\sigma_4=1$). Define the certificate

$$\widetilde R(n,\rho)\;:=\;\frac{\widehat X(n,\rho)}{4(n{+}3)(n{+}4)^2(2n{+}7)^2\,(\rho{-}n{-}1)(\rho{-}n{-}2)(\rho{-}n{-}3)}\,,\tag{13}$$

where $\widehat{X} \in \mathbb{Z}[n,\rho]$ is an explicit polynomial with $\deg_\rho \widehat{X} = 11$ and $\deg_n \widehat{X} = 20$, divisible by ρ^2; its full coefficient list ($\sim$180 integers) is embedded in the verification script (`code/verify_a269021.py`, table `XHAT`) rather than printed here. One anchor a reader can check by eye: the leading coefficient $[\rho^{11}]\,\widehat{X}$ equals $64n^{10}+1968n^9+26156n^8+198469n^7+952323n^6+3012795n^5+6333869n^4+8663374n^3+7264534n^2+3266000n+549760$ — precisely the degree-10 factor of $c_0(n)$.

Proposition 7 (the two identities)**.** *As identities of rational functions,*

$$\sum_{i=0}^{4} c_i(n)\,\sigma_i(n,\rho) \;=\; \widetilde{R}(n,\rho{+}1)\,\tilde{r}(n,\rho) \;-\; \widetilde{R}(n,\rho) \qquad \text{in } \mathbb{Q}(n,\rho), \qquad (\diamondsuit)$$

$$\sum_{c=0}^{3}\sum_{i=0}^{4} c_i(n)\,\frac{t(n{+}i,n{+}c)}{t(n{+}4,n{+}4)} \;=\; \widetilde{R}(n,n{+}4) \qquad \text{in } \mathbb{Q}(n), \qquad (\clubsuit)$$

where in ($\clubsuit$) each ratio $t(n{+}i,n{+}c)/t(n{+}4,n{+}4)$ denotes its closed rational form in n (fixed factorial offsets; spelled out in the script and guarded there against direct factorial values).

Proof. Each side of ($\diamondsuit$) is a rational function; clearing the explicit product of all denominators turns the difference into a polynomial $E(n,\rho)$. The structural bidegrees of the seven addends of ($\diamondsuit$), read off the factored forms (11)–(13) (recall $\deg c_i \le 21$, $\deg_n \widehat{X} = 20$, $\deg_\rho \widehat{X} = 11$):

addend	$\deg_n$ num.	$\deg_\rho$ num.	$\deg_n$ den.	$\deg_\rho$ den.
$c_i(n)\,\sigma_i(n,\rho)$, $0 \le i \le 4$	$21+3(4{-}i)$	$3(4{-}i)$	$5(4{-}i)$	$4{-}i$
$\widetilde{R}(n,\rho{+}1)\,\tilde{r}(n,\rho)$	21	14	10	7
$\widetilde{R}(n,\rho)$	20	11	8	3

The denominator degrees total 68 in n and 20 in ρ; each addend contributes to E at most its numerator degree plus the total denominator degree of the *other* addends, so $B_N = \max_t(\deg_n \mathrm{num}_t + 68 - \deg_n \mathrm{den}_t) = 89$ (attained at $c_4\sigma_4$) and $B_R = 28$ (attained at $c_0\sigma_0$ and at $\widetilde{R}$); the same bookkeeping is recomputed programmatically in the script (function `bound_from_terms`). Thus $\deg_n E \le 89$ and $\deg_\rho E \le 28$. A nonzero polynomial with $\deg_n \le B_N$, $\deg_\rho \le B_R$ cannot vanish on a grid of $(B_N{+}1)\times(B_R{+}1)$ points: fixing any grid row $n = n_0$ gives a univariate polynomial in ρ of degree $\le B_R$ vanishing at $B_R{+}1$ points, hence identically; then each coefficient, a polynomial in n of degree $\le B_N$, vanishes at $B_N{+}1$ points. We evaluate E exactly (rational arithmetic) at the 90×29 grid $n \in \{500,\dots,589\}$, $\rho \in \{5000,\dots,5028\}$, whose points avoid every zero of every denominator (all denominator factors are of the forms $\rho-n-k$, $2n+k-\rho$, $\rho+n+k$, $\rho+1$, $n+k$, $2n+k$ with $|k| \le 9$, none of which vanish there); all 2610 values are 0. Identity ($\clubsuit$) is proved the same way in one variable, with the structural degree bound $B_J = 351$ and 352 evaluation points. Both computations are part of the one-command verification (Section 4) and complete in seconds. $\square$

Remark 8. This grid argument is a deterministic identity test, not a probabilistic one: the degree bounds are explicit and the grid strictly exceeds them, so the vanishing forces the polynomial to be zero. It replaces a symbolic normalization of rational functions whose expanded numerators have hundreds of terms; nothing about it is numerical in the floating-point sense.

3.3 Telescoping

Lemma 9 (faithfulness at lattice points)**.** *Let $n \ge 1$ and $\rho \ge n+4$ be integers. Then: (a) no denominator appearing in σ_i, $\tilde{r}$, $\widetilde{R}(n,\rho)$ or $\widetilde{R}(n,\rho{+}1)$ vanishes; (b) $\sigma_i(n,\rho)\,t(n{+}4,\rho) = t(n{+}i,\rho)$ for all $0 \le i \le 4$; (c) $\tilde{r}(n,\rho)\,t(n{+}4,\rho) = t(n{+}4,\rho{+}1)$.*

Proof. (a) For $\rho \geq n+4$: the factors $\rho - n - k$ $(k \leq 3)$ are ≥ 1; $\rho + n + k$, $\rho + 1$, $2n + 2k + 1$, $2n + 2k + 2$, $n + 3$, $n + 4$, $2n + 7$ are positive; and in $\widetilde{R}(n, \rho{+}1)$ the shifted factors $\rho - n$, $\rho - n - 1$, $\rho - n - 2$ are ≥ 2. (b) On $n + 4 \leq \rho \leq 2n + 2i$, where both terms are nonzero, this is Lemma 6 iterated. For $2n + 2i < \rho \leq 2n + 8$ (where $t(n{+}i, \rho) = 0$ but $t(n{+}4, \rho) \neq 0$): the numerator of σ_i contains the factor $\prod_{k=i}^{3}(2n{+}2k{+}2{-}\rho)(2n{+}2k{+}1{-}\rho) = \prod_{s=2i+1}^{8}(2n{+}s{-}\rho)$, which vanishes at exactly those integer ρ — so both sides are 0, as *polynomial* vanishing, with no limit taken. For $\rho > 2n + 8$ both terms vanish and σ_i is finite by (a). (c) Same argument: for $n + 4 \leq \rho < 2n + 8$ it is Lemma 6; at $\rho = 2n + 8$ the factor $(2n{+}8{-}\rho)$ makes the left side $0 = t(n{+}4, 2n{+}9)$; for $\rho > 2n + 8$ both sides vanish. □

Theorem 10. *For every integer $n \geq 5$, $\sum_{\rho \in \mathbb{Z}}(Lt)(n, \rho) = 0$; that is, $(La)(n) = 0$.*

Proof. Define $G^*(n, \rho)$ for integer ρ by

$$G^*(n,\rho) \;=\; \begin{cases} 0, & \rho \leq n, \\ \displaystyle\sum_{q=n}^{\rho-1}(Lt)(n,q), & n+1 \leq \rho \leq n+3, \\ \widetilde{R}(n,\rho)\, t(n{+}4,\rho), & \rho \geq n+4. \end{cases}$$

We claim $(Lt)(n, \rho) = G^*(n, \rho{+}1) - G^*(n, \rho)$ for *every* integer ρ. For $\rho \leq n - 1$ both sides are 0 ($t(n{+}i, \cdot)$ is supported on $[n{+}i, 2n{+}2i]$, and $n \geq 5$ keeps all cumulative indices in range). For $\rho \in \{n, n{+}1, n{+}2\}$ it is the definition of the cumulative branch. For $\rho = n + 3$ it reads $\sum_{q=n}^{n+3}(Lt)(n, q) = \widetilde{R}(n, n{+}4)\, t(n{+}4, n{+}4)$, which is (♣) multiplied by $t(n{+}4, n{+}4) \neq 0$; here each closed form in (♣) equals the true ratio $t(n{+}i, n{+}c)/t(n{+}4, n{+}4)$ for every integer $n \geq 5$ by the same fixed-offset factorial manipulations as Lemma 6 (all factorial arguments are nonnegative for $n \geq 5$, and the vanishing cases $c < i$ are encoded by the empty binomial; the script additionally spot-checks all twenty forms against direct factorial values). For $\rho \geq n + 4$, multiply (♢) by $t(n{+}4, \rho)$ and apply Lemma 9: the left side becomes $(Lt)(n, \rho)$ and the right side $\widetilde{R}(n, \rho{+}1)t(n{+}4, \rho{+}1) - \widetilde{R}(n, \rho)t(n{+}4, \rho) = G^*(n, \rho{+}1) - G^*(n, \rho)$. Finally $G^*(n, \rho) = 0$ for $\rho > 2n + 8$ (there $t(n{+}4, \rho) = 0$ while $\widetilde{R}$ is finite by Lemma 9(a), since $\rho > 2n + 8 > n + 3$ keeps its denominator nonzero). Summing the telescoped identity over the finite support: $\sum_{\rho \in \mathbb{Z}}(Lt)(n, \rho) = \lim_{\rho \to \infty} G^* - \lim_{\rho \to -\infty} G^* = 0 - 0$. By (2), $\sum_\rho (Lt)(n, \rho) = \sum_i c_i(n)a(n{+}i) = (La)(n)$. □

Proof of Theorem 1. Theorem 10 gives $(La)(n) = 0$ for $n \geq 5$. For $0 \leq n \leq 4$ the relation is a finite integer identity, verified directly on the values computed from (2) (the script checks $n = 0, \ldots, 52$, of which only $n \leq 4$ is needed here; the research record checked $n \leq 80$ against an independent Toeplitz-determinant engine). Here $a(0) = 1$ — the empty permutation vacuously contains the empty pattern — and (2) also evaluates to 1 at $n = 0$ under the standard conventions, although Proposition 3 is stated for $n \geq 1$. This proves (1) for all $n \geq 0$. For the equivalence with the $\tilde{a}$-form of [1]: substituting $\tilde{a}_{n+i} = a(n{+}i)/(2n{+}2i)!^2$ into their Conjecture 17 and multiplying by $(2n{+}8)!^2$ yields exactly $4(n{+}3)(n{+}4)^2(2n{+}7)^2$ times (1) (a polynomial-coefficient identity checked in the script at points beyond its degree bound); the factor is positive for $n \geq 0$, so the two statements are equivalent. □

4 Validation and reproducibility

Everything is re-checkable with one command and no external data:

```
python code/verify_a269021.py     # expect "A269021 VERIFICATION: ALL 12 CHECKS PASS"
```

The script is Python 3 standard library only (observed: about 10 seconds under CPython 3.12 on a consumer desktop). Its twelve checks, in order: brute force over *all* $(2n)!$ permutations for $n \le 3$ against (2); the hook-length sum (3) for $n \le 10$; the 16 OEIS-listed terms; L annihilating the representation values for $n \le 52$ (of which only $n \le 4$ is *required* by the proof of Theorem 1 — the rest is extra validation); the ratio closed forms of Lemma 6 against direct factorial ratios; identity ($\diamondsuit$) on its 90×29 grid; the junction ratios against factorial values; identity ($\clubsuit$) on its 352-point grid; exhaustive telescoping ($(Lt)(n, \rho) = \Delta_\rho G^*$ at *every* integer ρ, and $\sum_\rho (Lt) = 0$, for $5 \le n \le 40$); and the equivalence with the $\tilde{a}$-form of [1].

The script is part of the proof object, not an accessory: the conjectured coefficients c_i and the certificate numerator $\widehat{X}$ are embedded in it verbatim, and the certificate is additionally shipped as the machine-readable file `code/certificate_xhat.json` (identical data). The empirical OEIS terms are used *only* as anchors, never as proof input. For reference, the SHA-256 digest of `verify_a269021.py` as distributed with this version of the paper is

`2dee0cdc613f6d96beaa10f6c2f9f915f3cd1f42e03c7cf25724adda13538042`

During the research phase the same material passed a larger battery: (2) against 81 independently computed terms, (8) against the exact LIS-distribution array for $n \le 26$, the grid identities at independently reconstructed certificates, and an independent automated audit that re-ran all computations and re-derived the small cases from separately written code. This paper's source and the verification script are publicly available at `https://github.com/alejlizardi/Periapsis_papers` (directory `a269021-kk-conjecture-17`); the fuller discovery record is available from the author on request.

5 Remarks and further work

Remark 11 (D-finiteness). As of [1], even the *existence* of some linear recurrence for $a(n)$ (equivalently for $\tilde{a}_n$) was open — they classify the sequence in their weakest tier, with neither the guessed recurrence nor D-finiteness proved, and we are not aware of a proof having appeared since. Representation (2) settles existence a priori: a finite sum of a proper hypergeometric term is P-recursive by the general Wilf–Zeilberger theory [6], independently of the particular certificate exhibited here.

Remark 12 (the coupled-diagonal regime). The obstruction named in [1] — fixed-pattern D-finiteness says nothing once the pattern length is coupled to the permutation length — is genuine: for the two-variable array $u_k(m) = \#\{\sigma \in S_m : \mathrm{LIS}(\sigma) \le k\}$, the fixed-$k$ Gessel determinant [4] has size growing with k, so fixed-pattern D-finiteness by itself provides no bounded finite representation along a coupled diagonal. What makes the diagonal $k \approx m/2$ accessible is Lemma 2: in the upper-half regime the determinantal gap expansion collapses to its one-point term, producing a *single* hypergeometric sum. The same collapse gives (8) for the whole upper half of the LIS distribution, and two-row/two-point truncations of the same kind may make neighboring diagonals (e.g. LIS $\ge n-1$ on S_{2n}, where at most two rows can be long) similarly finite — we have not pursued this.

Remark 13 (the operator's other solutions). Kauers and Koutschan observe that the recurrence has the hypergeometric term $(n-1)(2n)!$ among its solutions. This is consistent with, and independent of, Theorem 1: the order-4 operator L has a four-dimensional solution space, containing both that hypergeometric solution and $a(n)$ itself. The two are genuinely different solutions: by Kotesovec's asymptotic recorded on the OEIS entry, $a(n) \sim 16^n(n-1)!/(\pi e^2)$, so $a(n)/(2n)! \to 0$ super-exponentially — permutations of $[2n]$ with an increasing n-subsequence are a vanishing fraction of S_{2n}, even though their count satisfies this small recurrence.

Acknowledgements

The sequence A269021 was contributed to the OEIS by Alois P. Heinz, with extended terms by Vaclav Kotesovec, whose asymptotic is also quoted on the entry; the conjecture proved here is due to Manuel Kauers and Christoph Koutschan [1], and this note exists because their paper stated precisely what was missing. We thank the OEIS Foundation for maintaining the encyclopedia. The certificate was discovered with the zeilberger package of Maxima [9] (F. Caruso's implementation) and verified independently as described above. This work was AI-assisted: the mathematics, code and manuscript were produced with Claude (Anthropic) operating autonomously under the Periapsis research architecture, with the verification designed so that no step rests on the assistant's (or any software's) say-so; the named human author directed the campaign and takes responsibility for the content.

A The Kauers–Koutschan operator

The coefficients of (1), verbatim from the OEIS entry (M. Kauers and C. Koutschan, Mar 01 2023), also embedded in the verification script:

$$\begin{aligned}
c_0(n) &= -64(1+n)^2(2+n)^2(3+n)(1+2n)^2(3+2n)^2(5+2n)^2\, p_0(n),\\
c_1(n) &= \quad 16(2+n)^2(3+n)(3+2n)^2(5+2n)^2\, p_1(n),\\
c_2(n) &= -4(3+n)(5+2n)^2\, p_2(n), \qquad c_3(n) = 2\, p_3(n),\\
c_4(n) &= -3(4+n)(8+3n)(10+3n)\, p_4(n),
\end{aligned}$$

where

$$\begin{aligned}
p_0 &= 549760 + 3266000n + 7264534n^2 + 8663374n^3 + 6333869n^4 + 3012795n^5\\
&\quad + 952323n^6 + 198469n^7 + 26156n^8 + 1968n^9 + 64n^{10},\\
p_1 &= -5543040 - 487964n + 78563984n^2 + 229526554n^3 + 325846005n^4 + 284054698n^5\\
&\quad + 165789363n^6 + 67385886n^7 + 19359535n^8 + 3917758n^9 + 545913n^{10}\\
&\quad + 49788n^{11} + 2672n^{12} + 64n^{13},\\
p_2 &= -250467360 - 946158512n - 562569136n^2 + 3271711596n^3 + 9313604242n^4\\
&\quad + 12741784568n^5 + 11118771121n^6 + 6753094929n^7 + 2966908118n^8\\
&\quad + 959068672n^9 + 228837227n^{10} + 39928763n^{11} + 4969164n^{12} + 419248n^{13}\\
&\quad + 21568n^{14} + 512n^{15},\\
p_3 &= -1300242000 - 6360099840n - 10812498240n^2 - 778719870n^3 + 27672902302n^4\\
&\quad + 55047935941n^5 + 60004107039n^6 + 43766004538n^7 + 22820793074n^8\\
&\quad + 8747566435n^9 + 2488583381n^{10} + 523718876n^{11} + 80260596n^{12}\\
&\quad + 8677944n^{13} + 624800n^{14} + 26752n^{15} + 512n^{16},\\
p_4 &= -15900 - 61278n - 68928n^2 + 48699n^3 + 204716n^4 + 233810n^5 + 143536n^6\\
&\quad + 52389n^7 + 11324n^8 + 1328n^9 + 64n^{10}.
\end{aligned}$$

The certificate numerator $\widehat{X}(n,\rho)$ of (13) has $\deg_\rho = 11$, $\deg_n = 20$, is divisible by ρ^2, and satisfies $[\rho^{11}]\widehat{X} = p_0$; its full integer coefficient table is the constant `XHAT` in `code/verify_a269021.py` and, in machine-readable form, the file `code/certificate_xhat.json` (identical data).

B Numerical anchors

$a(0),\ldots,a(8)$: 1, 2, 23, 588, 24553, 1438112, 108469917, 9996042284, 1086997811325. Values of the refined formula (8) at $n=2$: $\#\{\mathrm{LIS}=2\}=13$, $\#\{\mathrm{LIS}=3\}=9$, $\#\{\mathrm{LIS}=4\}=1$ (total $23=a(2)$, and indeed $13+9+1$ counts all of S_4 except the LIS $=1$ permutation 4321). The script re-derives all of these from scratch.

References

[1] M. Kauers and C. Koutschan. *Some D-finite and some possibly D-finite sequences in the OEIS.* Journal of Integer Sequences **26** (2023), Article 23.4.5. Preprint: arXiv:2303.02793**v2** (which matches the published text; v1 misprints Conjecture 17's normalization as $a_n/(2n)!$, corrected to $a_n/(2n)!^2$ in v2). (Conjecture 17 and the surrounding discussion of `A269021`; Table 1, status 'O'.)

[2] A. Borodin, A. Okounkov, and G. Olshanski. *Asymptotics of Plancherel measures for symmetric groups.* J. Amer. Math. Soc. **13** (2000), 481–515. Preprint: arXiv:math/9905032. (The poissonized measure M^θ and the descent set $\mathcal{D}(\lambda)=\{\lambda_i - i\}$, §1.2; the determinantal formula for the correlation functions of $\mathcal{D}$ with the discrete Bessel kernel, second theorem of the introduction — Theorem 2; the kernel series used here, Proposition 2.6, eq. (2.14); an equivalent expansion $\sum_{s\geq 1} J_{x+s}J_{y+s}$ is also derived there. Statement numbers follow arXiv:math/9905032**v2**; the earlier arXiv version numbers the series proposition 2.7.)

[3] C. Schensted. *Longest increasing and decreasing subsequences.* Canad. J. Math. **13** (1961), 179–191. (The correspondence $S_m \to \bigcup_{\lambda\vdash m} \mathrm{SYT}(\lambda)^2$ with $\mathrm{LIS}(\sigma)=\lambda_1$; used as (3).)

[4] I. M. Gessel. *Symmetric functions and P-recursiveness.* J. Combin. Theory Ser. A **53** (1990), 257–285. (Fixed-k D-finiteness and the Bessel–Toeplitz determinant; background.)

[5] D. Zeilberger. *The method of creative telescoping.* J. Symbolic Comput. **11** (1991), 195–204. (The telescoper-plus-certificate proof pattern instantiated in Section 3.)

[6] H. S. Wilf and D. Zeilberger. *An algorithmic proof theory for hypergeometric (ordinary and "q") multisum/integral identities.* Invent. Math. **108** (1992), 575–633. (Proper hypergeometric terms are holonomic; existence half of the argument.)

[7] M. Petkovšek, H. S. Wilf, and D. Zeilberger. *A=B.* A K Peters, 1996. (General reference for Zeilberger's algorithm and WZ certification.)

[8] OEIS Foundation Inc. *Sequence `A269021`* (and `A269042`, `A047874`). The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A269021`. Sequence contributed by A. P. Heinz (2016); b-file terms 0..41 by A. P. Heinz and V. Kotesovec; "Conjectured recurrence" formula line by M. Kauers and C. Koutschan, Mar 01 2023 (entry version #30, unchanged as of 2026-07-01).

[9] Maxima Development Team. *Maxima, a Computer Algebra System* (version 5.46.0), with the `zeilberger` package by F. Caruso. `https://maxima.sourceforge.io/`. (Used for discovery only; the certificate it returned is proved independently in Section 3.)